

Module 1 Lesson 5
Real Numbers and Their Properties; Clock Addition (5.5)

Objectives

In this lesson, we will learn how to

- recognize the subsets of the real numbers;
- recognize the properties of real numbers;
- apply properties of real numbers to clock addition.

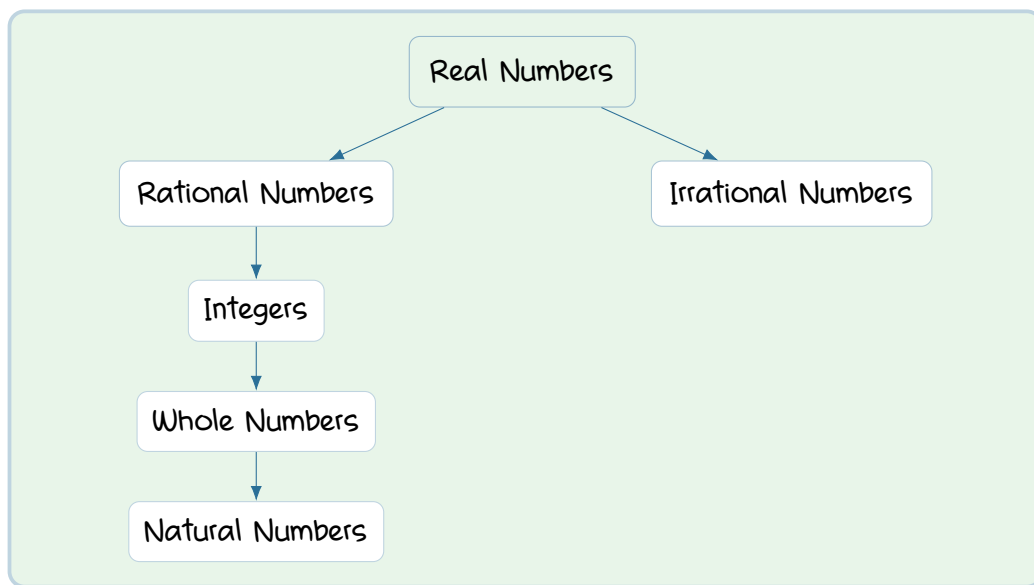
1. The Set of Real Numbers

The union of the rational numbers and the irrational numbers is the set of real numbers.

The sets that make up the real numbers are called subsets of the real numbers.

Subsets of the Real Numbers

$$\text{Real Numbers} = \text{Rational Numbers} \cup \text{Irrational Numbers}$$



2. Classifying Real Numbers

Common Number Sets

- **Natural numbers:** $1, 2, 3, 4, \dots$
- **Whole numbers:** $0, 1, 2, 3, 4, \dots$
- **Integers:** $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$
- **Rational numbers:** numbers that can be written as $\frac{a}{b}$, where a and b are integers and $b \neq 0$.
- **Irrational numbers:** numbers that cannot be written as a quotient of two integers. Their decimals do not terminate or repeat.
- **Real numbers:** all rational and irrational numbers.

Example 1: Consider the following set of numbers:

$$\left\{ 0, \sqrt{81}, -7, -\frac{3}{4}, 0.6, \sqrt{5}, 7.3, \pi \right\}.$$

List the numbers in the set that are

- | | |
|---------------------|------------------------|
| (a) natural numbers | (d) rational numbers |
| (b) whole numbers | (e) irrational numbers |
| (c) integers | (f) real numbers |

Solution

First notice that

$$\sqrt{81} = 9.$$

- (a) Natural numbers: $\sqrt{81}$, because $\sqrt{81} = 9$.
- (b) Whole numbers: $0, \sqrt{81}$.
- (c) Integers: $0, \sqrt{81}, -7$.
- (d) Rational numbers: $0, \sqrt{81}, -7, -\frac{3}{4}, 0.6, 7.3$.
- (e) Irrational numbers: $\sqrt{5}, \pi$.
- (f) Real numbers: $0, \sqrt{81}, -7, -\frac{3}{4}, 0.6, \sqrt{5}, 7.3, \pi$.

3. Properties of the Real Numbers

Important Properties

Let a , b , and c be real numbers.

Property	Addition	Multiplication
Closure	$a + b$ is a real number.	ab is a real number.
Commutative	$a + b = b + a$	$ab = ba$
Associative	$(a + b) + c = a + (b + c)$	$(ab)c = a(bc)$
Identity	$a + 0 = a$	$a \cdot 1 = a$
Inverse	$a + (-a) = 0$	$a \cdot \frac{1}{a} = 1, a \neq 0$

Distributive Property

Multiplication distributes over addition:

$$a(b + c) = ab + ac.$$

For example,

$$3(4 + 5) = 3 \cdot 4 + 3 \cdot 5.$$

Example 2: Name the property illustrated by each statement.

(a) $(4 + 7) + 6 = 4 + (7 + 6)$

(b) $8 \cdot 5 = 5 \cdot 8$

(c) $3(x + 2) = 3x + 6$

(d) $9 + (-9) = 0$

Solution

- (a) Associative property of addition.
- (b) Commutative property of multiplication.
- (c) Distributive property of multiplication over addition.
- (d) Inverse property of addition.

4. Rotational Symmetry

A **symmetry** of an object is a motion that moves the object back onto itself. After the motion, the object looks unchanged.

If it takes m equal turns to restore an object to its original position, and each turn gives a figure identical to the original figure, then the object has **m -fold rotational symmetry**.

Example 3: A regular hexagon can be rotated onto itself in six equal turns. What type of rotational symmetry does it have?

Solution

A regular hexagon has 6-fold rotational symmetry.

5. Clock Arithmetic

Clock addition is defined by adding numbers as if we are moving around a clock.

For a 6-hour clock, the possible values are

$$\{0, 1, 2, 3, 4, 5\}.$$

The symbol $\oplus$ is used for clock addition.

For example, on a 6-hour clock,

$$4 \oplus 3 = 1,$$

because starting at 4 and moving forward 3 hours lands on 1.

6-Hour Clock Addition Table

$\oplus$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

6. Properties in the 6-Hour Clock System

Example 4: Use the 6-hour clock system $\{0, 1, 2, 3, 4, 5\}$ with clock addition.

- (a) How can you tell that the set is closed under clock addition?
- (b) Verify one case of the associative property:

$$(2 \oplus 3) \oplus 4 = 2 \oplus (3 \oplus 4).$$

- (c) What is the identity element?
- (d) Find the inverse of each element.
- (e) Verify two cases of the commutative property.

Solution

- (a) The set is closed because all table entries are in

$$\{0, 1, 2, 3, 4, 5\}.$$

(b) First compute the left side:

$$(2 \oplus 3) \oplus 4 = 5 \oplus 4 = 3.$$

Now compute the right side:

$$2 \oplus (3 \oplus 4) = 2 \oplus 1 = 3.$$

Since both sides equal 3, this case verifies the associative property.

(c) The identity element is 0, because adding 0 does not change any element:

$$a \oplus 0 = a.$$

(d) The inverse of each element is the number that must be added to get 0:

Element	0	1	2	3	4	5
Inverse	0	5	4	3	2	1

(e) Two examples of the commutative property are

$$2 \oplus 5 = 1 \quad \text{and} \quad 5 \oplus 2 = 1,$$

so

$$2 \oplus 5 = 5 \oplus 2.$$

Also,

$$3 \oplus 4 = 1 \quad \text{and} \quad 4 \oplus 3 = 1,$$

so

$$3 \oplus 4 = 4 \oplus 3.$$

Summary

Key Ideas

- Real numbers consist of rational numbers and irrational numbers.
- Natural numbers, whole numbers, integers, rational numbers, and irrational numbers are all subsets related to the real number system.
- The real numbers satisfy important properties such as closure, commutative, associative, identity, inverse, and distributive properties.
- Clock addition is addition performed by moving around a clock.
- In a 6-hour clock system, every result of clock addition is in $\{0, 1, 2, 3, 4, 5\}$.
- The identity element for 6-hour clock addition is 0.